Second-Order ADMM with Augmented Newton Steps: Supplementary Document

1 Derivation of Hessian of StVK and Neo-Hookean Energy Densities

1.1 Saint–Venant Kirchhoff Energy Density

The expression for StVK energy density is as follows

$$\Psi_*^{stvk}(F) = \frac{\lambda_{\text{Lamé}}}{2}\left(\text{trace}\,(E)\right)^2 + \mu_{\text{Lamé}}\; E : E \tag{1}$$

where $E = 1/2\left(F^\top F - I\right)$ is Lagrangian finite strain tensor and F is deformation gradient [1]. The PK1 stress for the StVK energy density is

$$\frac{\partial \Psi_*^{stvk}(F)}{\partial F} = P(F) = 2\mu_{\text{Lamé}}\; E + \lambda_{\text{Lamé}}\; \text{trace}\,(E)\, I \tag{2}$$

Replacing Lagrangian finite strain tensor, E with infinitesimal linearized strain tensor $\boldsymbol{\epsilon} = 1/2(F + F^\top) - I$ and PK1 stress with linearized stress, $\boldsymbol{\sigma}$, we get

$$\boldsymbol{\sigma} = 2\mu_{\text{Lamé}}\; \boldsymbol{\epsilon} + \lambda_{\text{Lamé}}\; \text{trace}\,(\boldsymbol{\epsilon})\, I \tag{3}$$

Now, again differentiating the linearized stress, $\boldsymbol{\sigma}$, w.r.t. F

$$\frac{\partial^2 \Psi_*^{stvk}(F)}{\partial F^2} = (2\mu_{\text{Lamé}} + \lambda_{\text{Lamé}})\, I \tag{4}$$

1.2 Neo-Hookean Energy Density

The Neo-Hookean energy density can be expressed as

$$\Psi_*^{neo}(F) = \frac{\mu_{\text{Lamé}}}{2}\,(I_C - 3) - \mu_{\text{Lamé}} \log J + \frac{\lambda_{\text{Lamé}}}{2}(\log J)^2 \tag{5}$$

where $I_C = \text{trace}\left(F^\top F\right)$ and $J = \det(F)$.

The PK1 stress, $P(F) = \dfrac{\partial \Psi_*^{stvk}(F)}{\partial F}$, for the Neo-Hookean energy density is

$$\begin{aligned}
P(F) &= \mu_{\text{Lamé}} F - \mu_{\text{Lamé}} \frac{1}{J}\frac{\partial J}{\partial F} + \lambda_{\text{Lamé}} \frac{logJ}{J}\frac{\partial J}{\partial F} \\
&= \mu_{\text{Lamé}} F - \mu_{\text{Lamé}} \frac{1}{J} J F^{-\top} + \lambda_{\text{Lamé}} \frac{logJ}{J} J F^{-\top} \\
&= \left[\mu_{\text{Lamé}}\left(FF^\top - I\right) + \lambda_{\text{Lamé}} logJ\right] F^{-\top} \\
&\approx \mu_{\text{Lamé}}\left(FF^\top - I\right) + \lambda_{\text{Lamé}} logJ, \quad \text{using } F^{-\top} \approx I \\
&\approx 2\mu_{\text{Lamé}}\boldsymbol{\epsilon} + \lambda_{\text{Lamé}} logJ, \quad \text{using } FF^\top \approx I + 2\boldsymbol{\epsilon}
\end{aligned} \tag{6}$$

Finally, replacing $P(F)$ with linearized stress $\boldsymbol{\sigma}$ and using the approximations $J \approx 1 + \text{trace}(\boldsymbol{\epsilon})$ and $\log(1 + x) \approx x$, the final stress vector materializes as

$$\boldsymbol{\sigma} \approx 2\mu_{\text{Lamé}}\boldsymbol{\epsilon} + \lambda_{\text{Lamé}}\text{trace}(\boldsymbol{\epsilon})I \tag{7}$$

The stress tensor for the Neo-Hookean energy also resembles Equation 3 from the StVK energy. again differentiating the linearized stress, $\boldsymbol{\sigma}$, w.r.t. F yields $\partial^2 \Psi_*^{NH}(F)/\partial F^2 = (2\mu_{\text{Lamé}} + \lambda_{\text{Lamé}})\, I$ which is the same expression as Equation 4.

References

[1] Theodore Kim and David Eberle. 2022. Dynamic deformables: implementation and production practicalities (now with code!). In *ACM SIGGRAPH 2022 Courses* (Vancouver, British Columbia, Canada) *(SIGGRAPH '22)*. Association for Computing Machinery, New York, NY, USA, Article 7, 259 pages.